

Efficient multi-scale spatial aware 3D abdominal multi-organ segmentation model

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Abstract

Abdominal CT scans are naturally 3D images, and past models that only segmented 2D slices did not make good use of image contextual information, while processing 3D image data was computationally costly. In order to make full use of the context information of 3D images and reduce the computational cost it brings, we develop an efficient multi-scale spatial aware model for 3D abdominal multi-organ segmentation. Our proposed model consists mainly of an encoder, decoder, lightweight spatial attention and mixed pooling modules. We use 1D kernels and 2D kernels in the encoder and decoder modules to simulate a large 3D kernel for reducing the number of parameters and computational cost, while to better capture and exploit spatially contextual information, we use lightweight spatial attention and mixed pooling modules to capture cross-dimensional information, multi-scale features and long-range contextual information. We achieved a DSC of 89.86 and an NSD of 78.2 on the FLARE dataset. At the same time, the GPU occupation is only 1183MB, the parameter size is 9MB and the FLOPS is 335GB. Our model can achieve segmentation performance comparable to the best models with less number of parameters, lower memory usage and lower computational power.

Keywords: 3D Segmentation, Efficient Segmentation, Multi-scale, Abdominal Multi-Organ Segmentation

1 Introduction

The purpose of medical image segmentation is to make the anatomical or pathological structure changes in the image clearer. In recent years, due to the great improvement of its diagnostic efficiency and accuracy, it plays a key role in computer-aided diagnosis and intelligent medicine. Accurate and efficient automatic abdominal organ segmentation has important

clinical significance, which can reduce the manual participation of doctors and shorten the time of clinical diagnosis and treatment process. However, the diversity of abdominal CT data brings great challenges to automatic segmentation algorithms. Most of the existing abdominal CT datasets only contain single center, single-phase, single-device, or single-disease data, and algorithms designed based on this data show poor generalization performance in diverse clinical data. On the other

hand, since the CT image depicts the internal structure of the human body, it is a natural 3D image with natural spatial context information [1, 2], when slice-specific 2D models are applied to volumetric (3D) images (e.g., CT or MRI), it lacks spatial information and cross-plane contextual information. Using the 3D image segmentation model can obtain richer spatial information and context information than the 2D image segmentation model, however, the number of parameters and computing power required for 3D models is multiplied compared to 2D models, and the demand for computational resources and memory usage is greatly increased. In order to balance the consumption of computing resources and the accuracy of segmentation, most of the existing segmentation algorithms use the method of sliding window segmentation [7], which needs to increase the overlapping area of the segmentation, it also brings duplicate computation and resource consumption of image regions. Moreover, the separated image patches lose the global context information, which is not conducive to organ segmentation.

To this end, we developed an efficient and lightweight abdominal multi-organ segmentation model, based on the segmentation framework of U-Net [3], we also used four decoder modules and encoder modules. In order to increase the number of parameters and FLOPS as little as possible, we introduce Lightweight Spatial Attention (LSA) and Mixed Pooling Module (MPM) to more efficiently capture long and short distance context information and the space information between different features. Compared with the previous method of increasing the number of layers and sliding window segmentation, it significantly reduces GPU memory usage and time to generate segmented images.

In order to verify the generalization ability of the model on diverse datasets, we used the dataset from the FLARE (Fast and Low GPU memory Abdominal oRgan sEgmentation) competition, which includes kidney, pancreas, spleen, and liver segmentations. Moreover, the FLARE dataset has the characteristics of big amount of data, large distribution difference, multi-center, and multi-period equality, which can better evaluate the generalization performance of the model.

The main contributions of this paper are as follows: 1. A lightweight spatial attention module is

used to capture the spatial relationship of multiple organs through dimensional interaction. 2. Use the MPM module to capture long-distance and short-distance context information. 3. The results were achieved in the FLARE dataset with lower GPU usage, fewer parameters and FLOPS. The DSC is 89.86 and NSD is 78.2

2 Related Work

2.1 Medical Image Segmentation

Image semantic segmentation aims to achieve pixel classification of images. To this end, researchers propose an encoder-decoder structure, which is one of the most popular end-to-end structures, such as fully convolutional network (FCN) [5], U-Net [3], Deeplab[6], etc. In these structures, the encoder is usually used to extract image features, and the decoder is usually used to restore the extracted features to the original image size and output the final segmentation result. Most of the current deep learning research on medical image segmentation relies on the U-Net network and makes some changes to the network structure according to different tasks, but in fact non-network structure factors may also be important for improving the segmentation effect. Isensee et al. [7] believed that too much manual adjustment of the network structure may lead to overfitting to specific datasets, so they proposed a medical image segmentation framework called no new U-Net(nnU-Net), which can self-adapt any new dataset. nnU-Net automatically tunes all hyperparameters according to the properties of a given dataset without human intervention. Compared with 3D image segmentation, 2D image segmentation naturally lacks one-dimensional information, which sometimes limits the performance of the model.

2.2 3D Segmentation Model

Since medical data such as CT and MRI images mostly exist in the form of three-dimensional volume data, the use of three-dimensional convolution kernels can better mine the high-dimensional spatial correlation of data. Inspired by this, Çiçek et al. [14] extended the U-Net architecture to the application of 3D data, and proposed a 3D U-Net that directly processes 3D medical data. In

addition, Milletari et al. [15] proposed a similar architecture V-Net, which uses residual connections to avoid gradient disappearance, accelerate network convergence, and design a deeper network to achieve higher performance. Similarly, by applying residual connections to 3D networks, Yu et al. [16] proposed Voxresnet, Lee et al. [17] proposed 3D RU Net, and Xiao et al. [18] proposed Res-UNet. Although the amount of information that can be obtained from the 3D model has increased, the computing power requirements and model parameters have multiplied, which has a large computational cost burden in practical applications.

2.3 3D Segmentation Model Optimization

Most medical images are 3D volumetric data, but 2D convolutional neural networks cannot learn 3D spatial information, and 3D convolutional neural networks often require high computational cost and large GPU memory consumption. Based on 2D and 3D concatenation to optimize model complexity, Li et al. [8] proposed a Hybrid Densely Connected U-Net (H-DenseUNet), using a hybrid feature fusion layer to jointly optimize 2D and 3D features. Zhang et al. [9] proposed a lightweight hybrid convolutional network (LW-HCN) with a similar structure to H-DenseUNet, and the design of depth and space-time separation (DSTS) blocks further reduces the number of parameters and computational cost. Based on inverted bottleneck structure, He et al. [10] proposed that by compressing the number of kernels in the convolutional layers in the U-Net architecture, the model is forced to learn better feature representations while also reducing its size and computational complexity. Based on the use of asymmetric factorization, Szegedy et al. [11] proposed to decompose large 3D convolution kernels into multiple 1D convolution kernels. Adding attention and multi-scale pooling to capture spatial information more efficiently, the ACU-Net proposed by Hu et al. [12] includes a 3D spatial attention block to enrich the spatial details and feature representation of lesions in the decoding stage. Fan et al. [13] proposed to use a 3D-based hybrid pyramid pooling module to extract spatial context features and used inter-slice convolution and intra-slice convolution to further reduce the number of parameters.

Our model incorporates excellent 3D model compression methods such as the method of decomposing convolution reduces the amount of calculation. Additionally, we integrate a mixed pooling module that efficiently captures both spatial long and short distance information, as well as a lightweight spatial attention that effectively captures cross-dimensional information. By utilizing these techniques, we can replace the traditional methods of improving model performance through increased model width and depth or the use of second-order complexity attention.

3 Method

3.1 Overall Architecture

The architecture proposed in this paper uses a classic U-shaped structure as shown in Figure.1 and four layers of encoder blocks and decoder blocks are used. To improve spatial information capture, both Light Spatial Attention and Mixed Pool Module are employed.

Let the shape of the tensor sent to the encoder module be (C, H, W, D) . Similarly to the classic U-Net model, the number of channels is doubled every time an encoder block passes through, the H , W and D of the tensor all become $1/2$ of the original. The encoder block is mainly composed of two $3 \times 3 \times 3$ convolution and activation function relu . For the decoder block, the number of channels of each decoder block is halved, and the h , w and d of the tensor are doubled. In order to reduce the calculation time and the amount of model parameters, We divide the standard 3D convolution kernel with kernel size of $3 \times 3 \times 3$ into $3 \times 3 \times 1$ intra-slice convolution and $1 \times 1 \times 3$ inter-slice convolution, and then added to a $1 \times 1 \times 1$ convolution. A skip connection is used between each layer of encoder block and decoder block, and the number of channels between high pixels and low pixels is fitted, and element-wise addition of these. After the last decoder block, the number of channels is reduced to the number of organ types that need to be segmented through the convolutional layer.

3.2 Lightweight Spatial Attention

To capture cross-dimensional information and the spatial relationship between different features, we introduce lightweight spatial attention, as shown

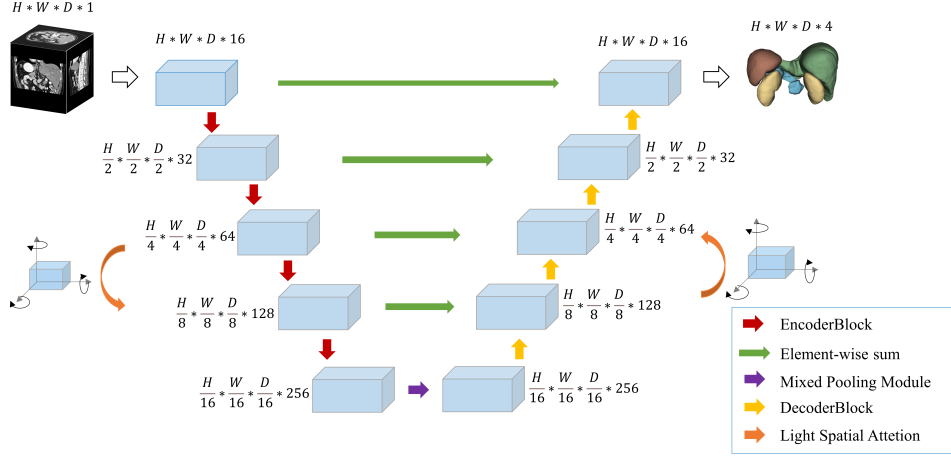


Fig. 1 Overview of the Model Architecture

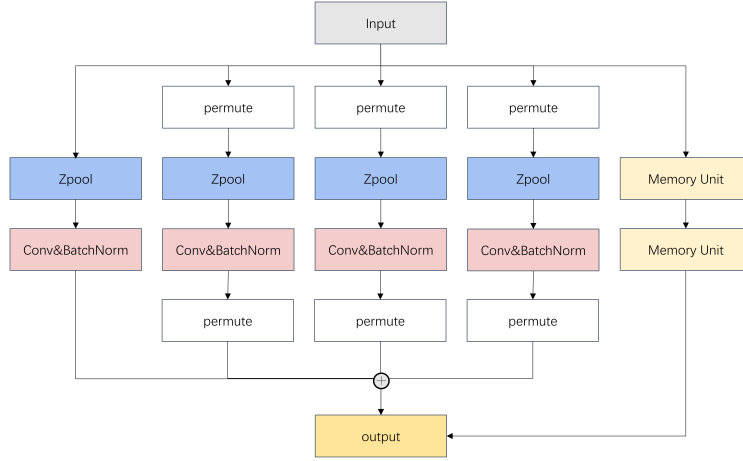


Fig. 2 This figure illustrates the lightweight spatial attention model, which includes dimension exchange calculation and memory unit information capture. The model aims to capture cross-dimensional information and spatial relationships between different features. In the dimension exchange calculation part, the input tensor dimensions are exchanged with the channel dimension to capture spatial relationships. After pooling, 3D convolution, and 3D instance normalization operations, the tensors are added together, and the average value is obtained.

in Figure.2. The model is mainly divided into two parts: dimension exchange calculation and memory unit information capture. In the first part, the main purpose is to exchange each dimension with the channel dimension, to capture the spatial relationship between different features. To generate a set of four different tensors, let $x \in \mathbb{R}^{C \times H \times W \times D}$ be the input tensor, the dimensions of $H, W,$ and

D are respectively exchanged with C . These tensors are (C, H, W, D) , (H, C, W, D) , (W, H, C, D) , and (D, H, W, C) . Subsequently, a pooling operation via $\text{Zpool}(1)$ is applied to each tensor. After pooling, 3D convolution and 3D instance normalization operations are performed on the tensors. These operations interact with both channel and spatial positions, resulting in calculations for each

position in the tensor. Next, the dimensions of the four tensors are restored to their original order, and they are added together. Finally, an average value is obtained by dividing the sum of the tensors by four.

This process results in a tensor with the same dimensions as the input tensor $\mathbf{x}$, but with each position in the tensor having been modified by the pooling, convolution, and normalization operations. Unlike other previous attentions that directly ignore the relationship in the channel dimension, we add the channel dimension to the convolution calculation, so that the channel dimension interacts with the spatial dimension and optimizes the model segmentation effect by using the relative positions between different features.

$$\mathbf{Z}_{\text{pool}} = [\text{MaxPool}_{0d}(\mathbf{x}), \text{AvgPool}_{0d}(\mathbf{x})] \quad (1)$$

where $\mathbf{x}$ represents the input tensor, and $0d$ represents the first dimension. The second part is to capture the global contextual information of the dataset by two memory cells to reduce the information loss due to the limited convolutional receptive field of the first part. After the tensor passes through the first memory cell, the number of channels is scaled to k . After normalizing the rows and columns separately, the number of channels is reduced through the second memory cell.

$$\mathbf{A} = \text{Norm}(\mathbf{F}\mathbf{M}_k^T) \quad (2)$$

$$\mathbf{F}_{\text{out}} = \mathbf{A}\mathbf{M}_v \quad (3)$$

where $\mathbf{A}$ is the attention matrix, $\mathbf{F}$ represents the input features, $\mathbf{M}_k$ and $\mathbf{M}_v$ represent two memory units respectively.

We sum these two parts with the original tensor to obtain the tensor enhanced by lightweight spatial attention. The memory unit makes up for the disadvantage that the convolutional layer can only capture information from neighbouring regions, bringing global contextual information. In terms of complexity, our lightweight spatial attention is reduced from the original quadratic complexity to primary complexity compared to the traditional attention, which uses only convolutional and linear layers in this paper, compared to the standard attention, greatly reducing FLOPS and improving the model prediction speed.

Compared with other previous attention mechanisms, our approach considers not only the

attention of the tensor in space, but also the attention between different channels. We optimize multi-organ segmentation by obtaining information across dimensions and spatial relationships between different features.

3.3 Mixed Pooling Module

The mixed pooling module is divided into two main parts, as shown in Figure.3. The first part mainly uses three different sized pooling layers to scale the input tensor to different sizes to obtain different sizes of perceptual field and refinement. Then it goes through 3D convolution with kernel size of 3 and instance normalization. The tensor output from the 3 branches is fused, and then a multi-scale feature-enhanced tensor is obtained by activation function and 3D convolution.

The second part passes tensors through strip pooling layers[4] and convolutions in three different dimensions to capture long-distance contextual information in each dimension. First, we input the tensor into three parallel paths, passing through the pooling layers of kernel size $(H, W, 1)$, $(H, 1, D)$ and $(1, W, D)$ respectively and reduce the tensor to one-dimensional.

$$\mathbf{y}_i^h = \frac{1}{W \times D} \sum_{0 \leq i < W} \sum_{0 \leq j < D} \mathbf{x}_{i,j,k} \quad (4)$$

where $\mathbf{y}_i^h$ represents the output of the input tensor $\mathbf{x}$ through the average pooling layer with kernel size $(H, W, 1)$, and $\mathbf{y}_i^w$ and $\mathbf{y}_i^d$ are generated identically to $\mathbf{y}_i^h$. Using 1D convolution to modulating the current location and its neighbour features, and then restore it to the original dimension by upsampling, just add the tensors of the three paths as follows:

$$\mathbf{y}_{c,i,j,k} = \mathbf{y}_{c,i}^h + \mathbf{y}_{c,j}^w + \mathbf{y}_{c,k}^d \quad (5)$$

where $\mathbf{y} \in R^{C \times H \times W \times D}$ is the output tensor, $\mathbf{y}^h \in R^{C \times H}$ and $\mathbf{y}^w \in R^{C \times W}$ and $\mathbf{y}^d \in R^{C \times D}$ are what we need to combine. Then we fuse them better by 3D convolution to get a long-range feature-enhanced tensor. The enhanced tensors of these two parts are spliced, and the convolution with a kernel size of 1 is used to reduce the number of channels to obtain the final output.

Multi-scale information is obtained through convolutional layers of different fineness, and the

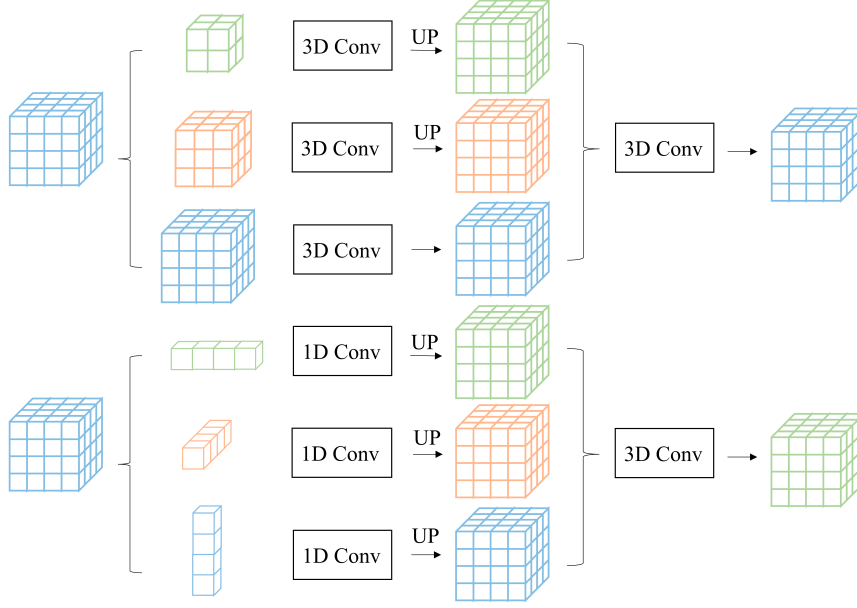


Fig. 3 Our mixed pooling module has two parts. The first part uses different sized pooling layers to refine the tensor, which is then fused to create a multi-scale feature-enhanced tensor. The second part captures long-distance contextual information in each dimension using strip pooling layers and convolutions.

long-distance dependencies between discretely distributed regions can be captured through the strip-shaped pooling layer and convolutional layer, and the two are combined so that the model can Capturing long-distance dependencies while paying attention to local details enhances the ability of the model to face 3D multi-organ segmentation in the complex environment of the abdomen.

4 Experiment

4.1 Dataset

Most existing abdominal datasets contain only single-center, single-stage, single-vendor, or single-disease cases, and models that achieve excellent performance in these datasets do not have definitive data to suggest that these excellent performances can be generalized to more diverse datasets and may only be applicable to more homogeneous and data format-demanding scenarios.

The FLARE dataset was derived from the FLARE abdominal multi-organ segmentation competition. Part of the dataset was adapted from licensed datasets from MSD [19] (liver [20], spleen, pancreas), NIH Pancreas [21–23], and KiTS [24,

25], and also includes a test set consisting of 50 abdominal CT cases from Nanjing University. The abdominal CT training data are characterized by large distribution differences, multi-center and multi-phase as well as the inclusion of pathological organs, and include multi-organ segmentation labels for both liver, kidney, spleen and pancreas. The generalization ability of the model in the face of relatively large differences in CT image segmentation can be effectively tested.

There are a total of 361 CT images on the training set and a total of 50 CT images on the validation set of the competition. 5-fold cross-validation is used on the training set in order to improve the model performance.

4.2 Evaluation Metrics

This paper mainly focuses on the following four metrics: Dice Similarity Coefficient (DSC), Normalized Surface Dice (NSD)[26], GPU memory usage, model parameter amount and model FLOPS.

DSC and NSD are mainly used to evaluate the accuracy of model segmentation, and DSC is the main score. GPU memory usage, model parameters, and FLOPS are used to measure the GPU

resources and computing power required for model segmentation.

$$DSC = \frac{2|G \cap S|}{|G| + |S|} \quad (6)$$

where G it represents the ground truth and S represents the segmentation result.

$$NSD = \frac{|G \cap B_s| + |S \cap B_g|}{|G| + |S|} \quad (7)$$

B_s it represents the regional boundary of the prediction result within the allowable error range, and B_g represents the regional boundary of the basic facts within the allowable error range.

4.3 Comparison of Segmentation Results

After our experimental comparison, the average DSC of UNet is 73.9, nnU-Net is 82.64, and our average DSC is 89.86. Due to the significant performance gap seen in UNet’s results, we will primarily compare the segmentation results of nnU-Net, which is also the baseline model in the FLARE competition, nnU-Net outperformed most of the models customized for the competition in many open international competitions without manual tuning and achieved the best results in many tasks.

We calculated the DSC and NSD metrics using the official validation tool. Comparing the first line in Figure 4, we can see that the segmentation results of nnU-Net and our multi-organ segmentation are similar and achieve a high level of accuracy. Even when observed without specialized tools, the difference is negligible. From the perspective of quantitative metrics, on this CT image, the average DSC of nnU-Net and our multi-organ segmentation are 0.9616 and 0.9611, respectively. Both methods achieve an advanced level of segmentation.

However, when the location of the kidney and the pancreas are close, or when the kidney is diseased, the segmentation effect of nnU-Net is very unsatisfactory. As can be seen from the second row in the figure, the segmentation effects of the spleen and liver are relatively good. The nnU-Net and our DSC on the spleen segmentation are 0.9831 and 0.9815 respectively, and the nnU-Net and our DSC

Table 1 DSC comparison with nnU-Net

	nnU-Net	Our
Mean DSC	82.64	89.86
Liver DSC	95.51 ± 4.50	96.15 ± 4.60
Kidney DSC	80.73 ± 18.07	93.27 ± 8.25
Spleen DSC	87.68 ± 23.52	96.20 ± 2.96
Pancreas DSC	66.63 ± 21.61	73.81 ± 18.84

on the liver segmentation are 0.9822 and 0.9886 respectively, but in the segmentation of kidney and pancreas, nnU-Net segmentation of the two is blurred, and there are a lot of misjudgments. The DSC of nnU-Net for kidney and pancreas is 0.7878 and 0.3510, and the performance is greatly reduced, and we can observe significant overlap in the segmentation of the kidney and pancreas in the results of the third row segmentation result.

Our model can still maintain a high segmentation performance for such organs that are relatively similar and for organ lesions. In the second row segmentation result in the figure, our model can effectively identify the spatial location of the kidney and pancreas, and the DSC of the liver and pancreas are 0.9754 and 0.8825, respectively, much higher than the results of nnU-Net. In the third row segmentation result, in the case of kidney lesions, the model in this paper can still identify the kidney and pancreas well, with DSC of 0.9512 and 0.8005, respectively.

From the comparison of the segmentation results, it can be seen that when the organ position and size are relatively stable, the segmentation performance of this model and nnU-Net is not much different, but when the organ size and position are not stable or are relatively close to other organs In the case of nnU-Net, there will be a large misjudgment situation, but the model in this paper still performs relatively well, with good stability and generalization ability.

4.4 Comparison of Quantitative Metrics

Table 1 and Table 2 compare the overall segmentation performance of our model and nnU-Net in the FLARE competition validation set. It can be seen from Table 1 that the model in this paper exceeds nnU-Net by 6.22% on average DSC, and is higher than nnU-Net on all individual segmentations, and the similar variance is much higher than

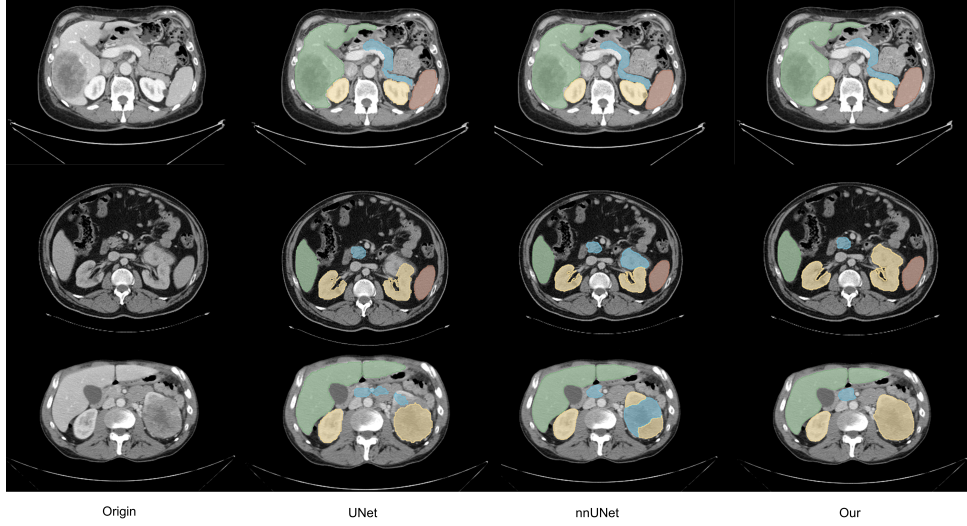


Fig. 4 Comparison of Segmentation Results. From left to right, the images represent the original image, segmentation results of UNet, segmentation results of nnU-Net, and our results.

Table 2 NSD comparison with nnU-Net

	nnU-Net	Our
Mean NSD	73.02	78.2
Liver NSD	80.98 ± 13.69	80.87 ± 12.81
Kidney NSD	72.52 ± 19.52	83.06 ± 13.17
Spleen NSD	81.72 ± 23.22	89.27 ± 10.58
Pancreas NSD	56.87 ± 17.82	59.59 ± 18.41

nnU-Net on organs segmentation except for liver., indicating that the model in this paper encounters other abnormal situations such as similar organs or lesions, and most of the organ segmentation performance can still maintain the original level.

In the evaluation metrics of NSD, the model of this paper exceeds nnU-Net by 5.18%, and the result of liver segmentation is similar, and it is higher than nnU-Net by 2.72% in pancreas, but it is much higher than nnU Net in the NSD metric of kidney and spleen. The segmentation results are 12.54% and 9.52% respectively, and the variance of kidney and spleen is also small, which also reflects the strong generalization ability of the model. In addition to the metrics of DSC and NSD, we also compared the parameter amount and FLOPS of each competition model. It can be seen from Table.3 that our model has achieved the best results on DSC, surpassing EfficientSegNet, the model with the best comprehensive performance in FLARE, and the nnU-Net model with the best

DSC performance in FLARE, and the GPU occupancy, parameters The volume and FLOPS are only 1183MB, 9MB, and 335 GB, which belong to the first echelon results in the model comparison. The model in this paper uses 1/3 of the GPU occupancy, 1/7 of the parameters and 1/4 of the FLOPS of the highest DSC model to achieve comparable results to its DSC metric, the comparable performance with fewer parameters and arithmetic power.

4.5 Ablation Study

The ablation study is shown in Table.4. BaseNet is the basic network architecture that removes the Lightweight Spatial Attention (LSA) and Mixed Pooling Layer (MPM), i.e., the basic U-Net network and the encoder and decoder modules. The average DSC of the model on the FLARE dataset is 87.83, and the average NSD is 76.45. When only the mixed pooling module is added, the average DSC of the model is 89.65 and the average NSD is 77.85, indicating that the mixed pooling layer has greatly improved the model's performance. When only the lightweight spatial attention module is added, the average DSC of the model is 88.39 and the average NSD is 76.61, which shows that the lightweight spatial attention module also greatly improves the performance of the model. The average DSC and NSD of the model proposed in this

Table 3 The average DSC, GPU Memory, parameters and FLOPS comparison of each model, where nn-UNet (restructured) is to adjust the nn-UNet model feature maps to 48 and patch the optimal result with size $80 \times 192 \times 160$

model	mean DSC	GPU	parameters	FLOPS
U-Net [3]	73.9	1350MB	23MB	319 GB
nnU-Net(baseline) [7]	82.64	3131MB	62MB	562 GB
COBRA [26]	83.7	only CPU	1.6MB	57 GB
EfficientSegNet [13]	89.5	1177MB	9MB	333 GB
nnU-Net (restructured) [7]	89.85	3463MB	63MB	1.25 T
Our	89.86	1183MB	9MB	335 GB

Table 4 Ablation study. Note: BaseNet: the basic network architecture, LSA: Lightweight Spatial Attention, MPM: Mixed Pooling Layer

Method	Mean DSC	Mean NSD
BaseNet	87.83	76.45
BaseNet+MPM	89.65	77.85
BaseNet + LSA	88.39	76.61
BaseNet+ LSA+MPM	89.86	78.2

paper are the best results, which are 89.86 and 78.2, respectively.

5 Conclusion

The purpose of this research is to explore a 3D abdominal multi-organ segmentation model that is lightweight, has strong generalization ability and maintains high performance, to solve some defects in the application of existing models, such as high computational cost and poor generalization ability, when the model is actually applied in clinical settings, the diversity of the dataset can result in poor model performance, and the model’s size may be excessively large, requiring excessive hardware resources. This paper proposes a new model framework that adopts various optimization strategies to improve the performance of the model, such as a mixed pooling module that can capture multi-scale and long-distance information and a lightweight attention model that can capture cross-dimensional spatial information. In terms of model lightweight, we use the method of convolution decomposition, use convolution and pooling instead of attention to obtain long-distance information and spatial information, etc. The method proposed in this paper works well on healthy or slightly lesion-affected organs and has strong stability in the face of multi-phase, multi-center FLARE datasets. As seen from the DSC

and NSD metrics, our method excels on segmentation of the liver, kidney and spleen, achieving a high degree of generative power. Also with similar performance on DSC and NSD, there is a significant reduction in the number of parameters and FLOPS on the model, which greatly reduces the complexity of the model and training prediction time, using less resources, but the model performance can still remain in the first echelon. The information contained in 3D images is naturally more than that contained in 2D images, but it also increases the parameters of the model and the burden of computing resources, which may cause difficulties in practical applications. Our model can solve this problem very well. However, the training of the 3D fully supervised model relies on high-quality labeling data, and usually requires sufficient 3D image labeling. If the labeling data is incomplete or inaccurate, the performance of the model will also be affected, and the labeling of 3D images is relatively the cost of 2D images is a hundredfold. In the future, we will use semi-supervised or self-supervised methods, which can greatly reduce the requirements for datasets, and at the same time add pre-training methods to speed up model convergence, which is more suitable and convenient for training on new datasets.

Acknowledgments. This work is supported by National Natural Science Foundation of China, Grant No.11702289 and Key core technology and generic technology research and development project of Shanxi Province, No. 2020XXX013.

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